Assignment

Q2. Take one Domain and draw the graph (Normal Distribution) – Empirical Rule

Introduction

Normal distribution is a continuous probability distribution that is symmetric around its mean. It forms a bell-shaped curve where most of the data values are concentrated near the average, and fewer values are found farther from the mean.

In many real-life situations such as delivery time, exam scores, height of people, and product demand, data follows a normal distribution pattern. The Empirical Rule (68–95–99.7 rule) helps us understand how data is distributed within standard deviation ranges.

Domain: Online Food Delivery Time

In this assignment, the domain selected is online food delivery time. The time taken to deliver food orders varies depending on distance, traffic, and restaurant preparation time. However, most deliveries are completed around an average time, forming a normal distribution pattern.

Mean and Standard Deviation

Assume that the average food delivery time is 30 minutes. This value represents the mean (μ). The variation in delivery time is measured using standard deviation (σ), which is assumed to be 5 minutes.

So,

- Mean (μ) = 30 minutes
- Standard Deviation (σ) = 5 minutes

This means most deliveries occur around 30 minutes, with some variation on either side.

Empirical Rule Calculations

Empirical Rule (68–95–99.7 Rule)

The empirical rule explains how data is distributed around the mean in a normal distribution.

- **$\mu \pm 1\sigma$ Range**

$\mu = 30$

$\sigma = 5$

So,

$30 - 5 = 25$

$30 + 5 = 35$

Range = **25 to 35 minutes**

➞ About **68%** of deliveries happen between 25 and 35 minutes.

- **$\mu \pm 2\sigma$ Range**

$30 \pm 2(5)$

30 ± 10

$30 - 10 = 20$

$30 + 10 = 40$

Range = **20 to 40 minutes**

➞ About **95%** of deliveries happen between 20 and 40 minutes.

- **$\mu \pm 3\sigma$ Range**

$30 \pm 3(5)$

30 ± 15

$30 - 15 = 15$

$30 + 15 = 45$

Range = **15 to 45 minutes**

➞ About **99.7%** of deliveries happen between 15 and 45 minutes.

Graph Representation (Normal Distribution)

The following graph represents the empirical rule for the selected domain.

Normal Distribution Curve Showing Empirical Rule

Interpretation

From the normal distribution graph, we observe that most food deliveries are completed close to the average time of 30 minutes. According to the empirical rule:

- Around 68% of deliveries are completed between 25 and 35 minutes.
- Around 95% of deliveries are completed between 20 and 40 minutes.
- Almost 99.7% of deliveries are completed between 15 and 45 minutes.

This shows that the delivery system is mostly consistent, and extreme delays are very rare.

If the standard deviation increases, it means delivery times are more spread out and less consistent. If it decreases, deliveries become more stable and predictable.

Conclusion

Normal distribution helps in understanding how real-world data is spread around the mean. In the case of food delivery time, the empirical rule allows us to measure consistency and performance. By analysing the mean and standard deviation, businesses can identify inefficiencies and improve service quality.

Thus, normal distribution and the empirical rule play an important role in statistical analysis and machine learning applications.